Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/11/2024
PI Name:	Didier, Kaylin Ph.D.		
Project Title:	Ionizing Radiation and Immune Responses: Exploring Sex Differences		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53715	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 TRISH-RFA-2101-PD: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	09/01/2021	**End Date:**	12/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2023 per TRISH (Ed., 8/5/23) NOTE: End date changed to 08/31/2023 per TRISH (Ed., 2/23/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schrage, William Ph.D. (MENTOR: University of Wisconsin, Madison)		
Grant/Contract No.:	NNX16AO69A-P0603		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Astronauts will soon be deploying on deep space missions to Mars, and will be exposed to continuous low dose, high energy ionizing radiation (IR). This IR is different from Earth orbit radiation; therefore, this kind of space radiation may lead to undefined acute and chronic health problems from continuous radiation exposure. Previous research on the effects of space radiation have predominantly studied male astronauts. Data from cancer patients receiving radiation, or animal research, suggest women might show an exaggerated inflammation response, making them more likely to develop diseases of the heart or blood vessels (vascular cells). These unknown sex differences remain as unexplored as deep space. The purpose of this fellowship proposal is to investigate how men and women differ in their immune and vascular response to IR, as radiation may increase or accelerate disease burden in astronauts who traveled into deep space. The first study goal is to test increasing levels of IR to see if the inflammation response is more sensitive in one sex versus the other. Immune cells from healthy adults will be exposed to IR and followed for 24 hours. Measurements include signals the cells produce, signals inside and on the surface of immune cells that change their function. In the second study, immune and vascular cells are placed together after IR, to see if crosstalk between cells is different between the sexes. Methods from study one will be repeated in both cell types, plus measuring adhesion molecules on the surface of vascular cells that allow immune cells to bind and amplify inflammation. Crosstalk is expected to magnify the inflammation response. The findings are expected to help develop sex-specific interventions to minimize risk from IR, so astronaut health and mission success are maximized.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To address several of the noted limitations from both clinical, animal, and cellular studies, and to make a substantial inroad into human pathophysiology of IR, our approach includes the following: 1) tightly controlling repeatable radiation type and dose rate relevant to space environment, 2) well-characterized healthy adult population, with robust sex hormone profiling, 3) quantifying numerous circulating biomarkers which focus on a key control points in vascular inflammation (S100A8/9, DAMPs, cytokines), and 4) immune-vascular interactions to interrogate a complex network of radiation damage and immunologic response. Moreover, our research team expertise encompasses immunology and cardiovascular physiology to maximize interpretation of data. In summary, our innovative research design and study team are poised to make a substantial leap forward into understanding sex-specific mechanisms of ionizing radiation damage.
Task Progress:	Aim 1 was to determine whether the immune cell response to acute ionizing proton radiation is sex dependent. We found that after 48 hours, 1Gy of proton radiation affects the anti-/pro-inflammatory response in a sex and cytokine specific manner. Females displayed greater anti-inflammatory cytokines (IL-10 and IL-4) compared to males after 1Gy of radiation. In the majority of pro-inflammatory cytokines measured there was no change in the expression of the cytokines nor sex differences. However, IL-2 was greater in females than males after 1Gy of radiation, but INF-gamma was increased in males compared to females. Aim 2 was to determine if the immune cells response to stimulus is impacted post acute ionizing proton radiation. We showed that 48 hours after 1Gy of radiation the expression of anti-/pro-inflammatory cytokines is increased in a similar manner between sexes in response to a stimulus. We also found that 72 hours after 1 Gy of radiation there was no sex difference in the response of anti-/pro-inflammatory cytokines. There was also a similar increase in stimulated cytokine expression after 72 hours 1Gy compared to 48 hours. However, we found that 72 hours post 1 Gy of radiation had a greater stimulated cytokine expression compared to 48 hours. Future studies include increasing our sample size, utilizing different dosages of proton radiation, performing repeated dosages of proton radiation, and using flow-cytometry to determine the intracellular immune and sex dependent response of macrophages. This project showed that immune cells that are exposed to proton radiation, which is the predominate form of space radiation, produce inflammatory markers in a sex specific manner. We also showed that the immune cells may have an exaggerated response to a stimulus after they have irradiated with proton radiation. This information indicates that the immune cells exposed to an acute dose radiation have a lower stimulus threshold. This can lead to an exaggerated immune response to a viral or bacterial stimulus. Countermeasures to the priming of the immune cells could be deployed or developed to prevent the immune cells from being overstimulated.
Bibliography Type:	Description: (Last Updated: 06/21/2024)
Abstracts for Journals and Proceedings	Didier KD, Sanchez S, Forsberg MH, Shahi A, Wannebo BM, Capitini CM, Schrage WG. "Inflammatory response to radiation exposure of immune cells." American Physiology Summit 2023, Long Beach, California, April 20-23, 2023. American Physiology Summit 2023 Meeting Abstracts. 2023 May 23;38(S1). https://doi.org/10.1152/physiol.2023.38.S1.5733308 , May-2023
Abstracts for Journals and Proceedings	Sanchez S, Didier KD, Forsberg MH, Shahi A, Wannebo BM, Capitini CM, Schrage WG. "TNF-alpha stimulated immune response: Are there sex differences?" American Physiology Summit 2023, Long Beach, California, April 20-23, 2023. American Physiology Summit 2023 Meeting Abstracts. 2023 May 23;38(S1). https://doi.org/10.1152/physiol.2023.38.S1.5733308 , May-2023
Abstracts for Journals and Proceedings	Didier KD, Sanchez S, Forsberg MH, Shahi A, Wannebo BM, Capitini CM, Schrage WG. "Inflammatory response to radiation exposure of immune cells: Exploring sex differences." 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. , Feb-2023
Articles in Peer-reviewed Journals	Didier KD, Hammer SM, Alexander AM, Rollins KS, Barstow TJ. "The acute effects of passive heating on endothelial function, muscle microvascular oxygen delivery, and expression of serum HSP90a." Microvasc Res. 2022 July;142:104356. https://doi.org/10.1016/j.mvr.2022.104356 ; PMID: 35276210 , Jul-2022

Articles in Peer-reviewed Journals	Didier K, Sanchez S, Shahi A, Capitini C, Schrage W. "Extracellular and intracellular inflammatory response of immune cells to radiation exposure." Physiology. 2024 May 21;39(S1):989. https://doi.org/10.1152/physiol.2024.39.S1.989 , May-2024
Awards	Didier K. "2024 Environmental & Exercise Physiology Section Dr. Charles (Tip) Tipton Postdoctoral Research Award, January 2024." Jan-2024
Awards	Didier K. "Rankin-Skatrud Lectureship Conference Stipend for Postdoctoral Fellow. January 2023." Jan-2023
Awards	Didier K. "University of Wisconsin – Madison Symposium Blitz Talk Judge's Choice Winner, September 2022. " Sep-2022
Awards	Sanchez S. "2023 CVRC Summer Undergraduate Fellowship, May 2023. " May-2023
Awards	Sanchez S. "American Physiological Society Summer Undergraduate Research Fellowship, March 2022. " Mar-2022